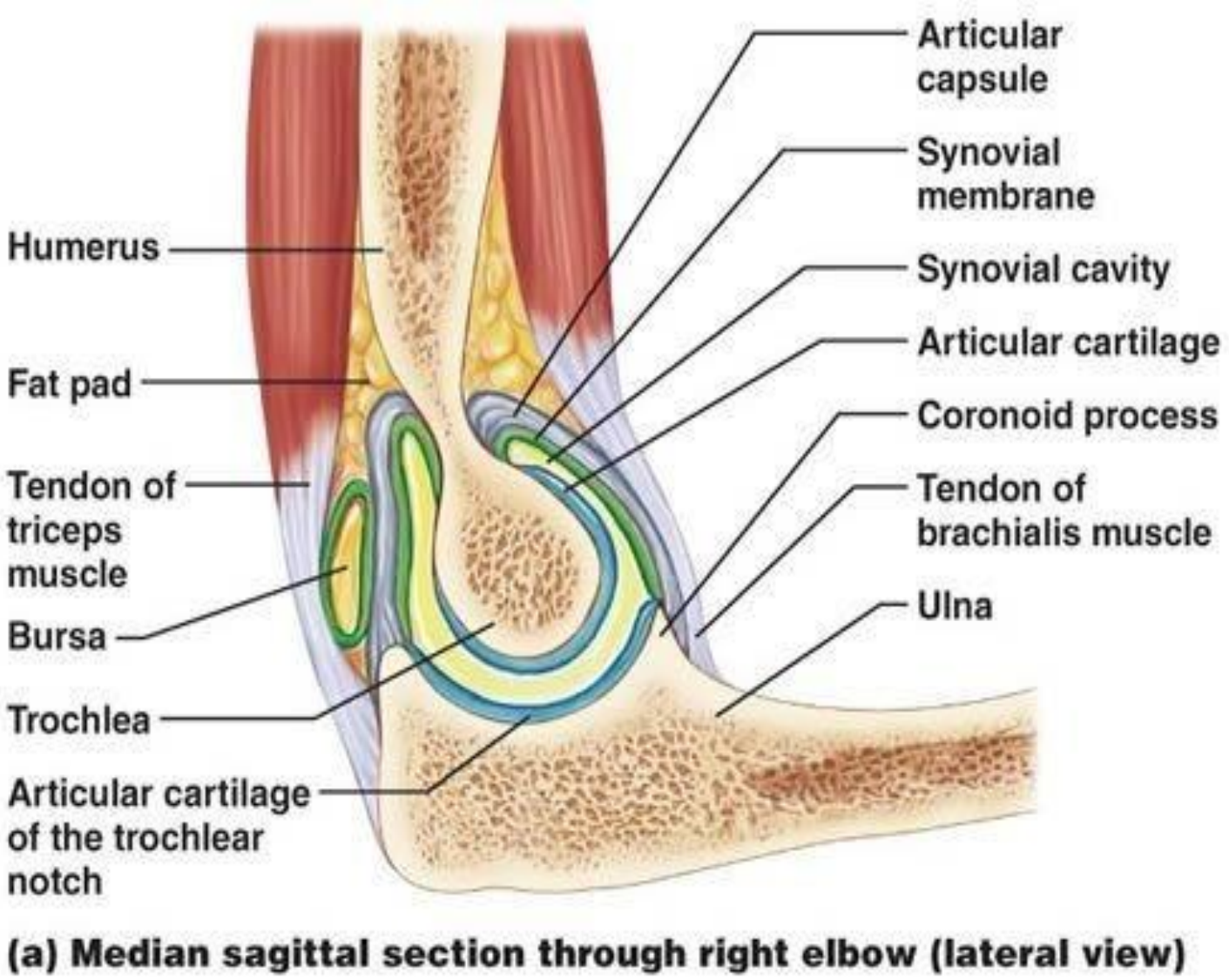
Elbow Joint Assessment

Walaa Abu-Taleb



Learning Outcomes

- **By the End of this lesson, you should be able to:**
- Practice MSK assessment relevant to the elbow joint.

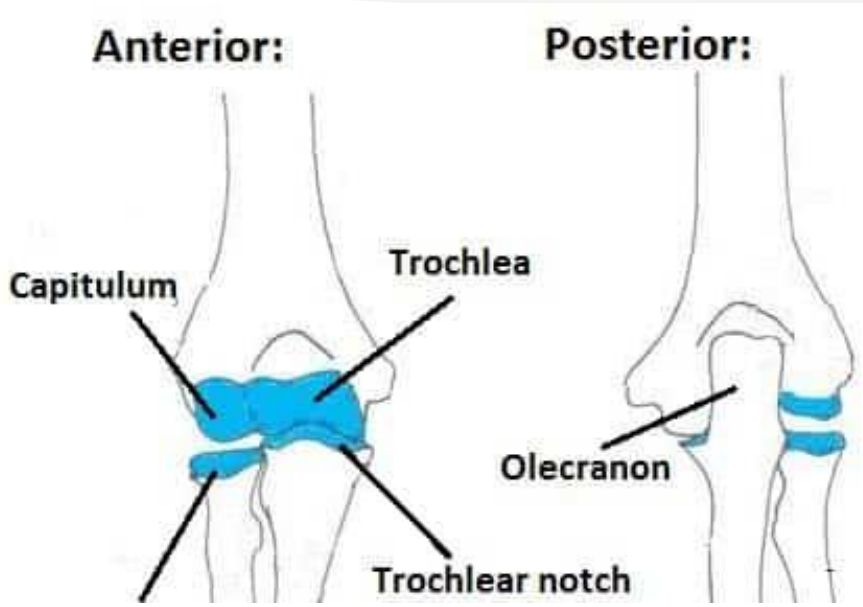


(a) Median sagittal section through right elbow (lateral view)

Elbow Joint

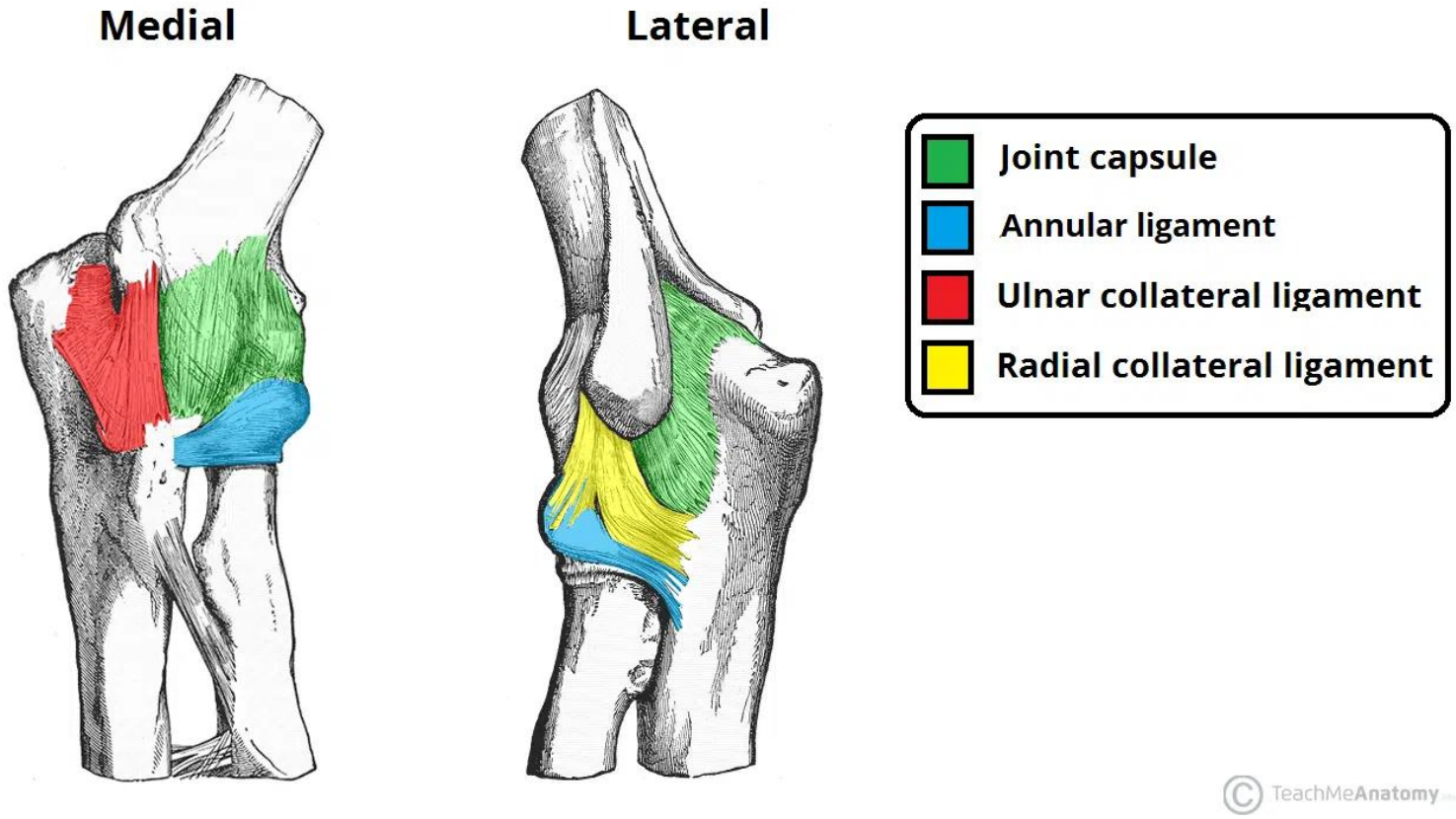
- The elbow joint consists of two separate articulations:

- Trochlear notch of the ulna and the trochlea of the humerus
- Head of the radius and the capitulum of the humerus
- Note: The proximal radioulnar joint is found within same joint capsule of the elbow, but most resources consider it as a separate articulation.





Ligaments


Medial
Lateral
Joint capsule
Annular ligament
Ulnar collateral ligament
Radial collateral ligament
© TeachMeAnatomy

Specific questions for the elbow joint

- Exact location of pain
- Timeline-When are the patients reported symptoms at their worst?
- Mechanism of the injury- In the case of a traumatic event, the mechanism of injury helps guide the diagnosis. For a traumatic injuries, specific symptoms can be highly useful in determining a diagnosis. For example: patient reported numbness and/or tingling in the 5th digit may suggest ulnar neuropathy.
- Presence of numbness or tingling?
- Do your symptoms change (better or worse) with any movements of the neck or shoulder?
- Does your elbow ever “slip out” or feel unstable?
- Does the pain change with gripping activities?
- Do you ever experience numbness of tingling in the hand?
- Was the elbow hyper extended during the time of injury?
- Do you relate the symptoms to a throwing activity?

Red Flags

- Fracture/Dislocation (FOOSH) (Positive Elbow Extension Test)
- Biceps/triceps tear
- Malignancy
- Abnormal Vascular Exam
- Heterotopic Ossification (Post-Surgical Consideration)
- Inappropriate progress from treatment made after surgery

Observation

General posture of the upper quarter: Proximal factors should be considered which could predispose the patient to elbow symptoms.

Thoracic and cervical spine including kyphosis and forward head

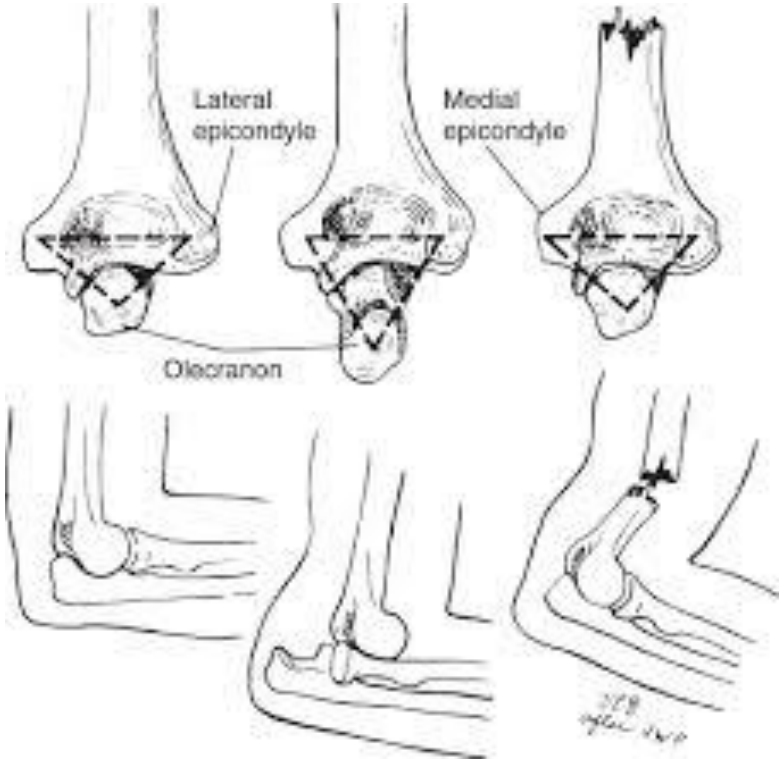
Carrying angle: The carrying angle has a mean value of 10 degrees for men and 13 degrees for women.[4] variability occurs till age of 14 -15

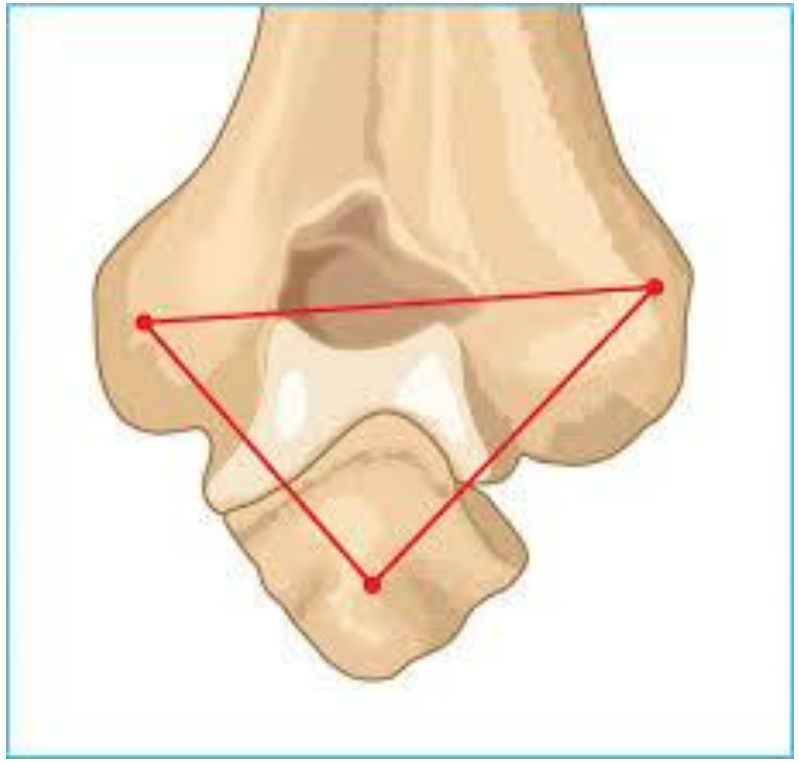
Swelling/ecchymosis/deformities/muscle wasting

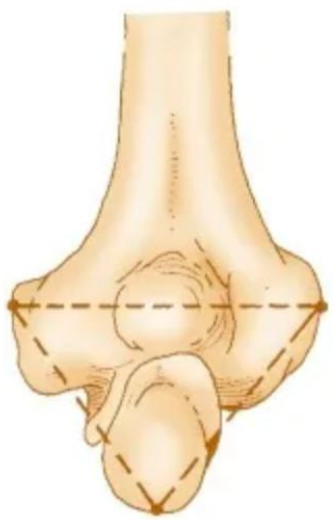
Triangle Sign

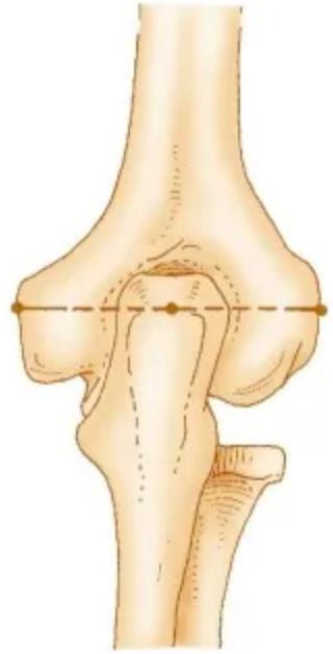


Triangle Sign

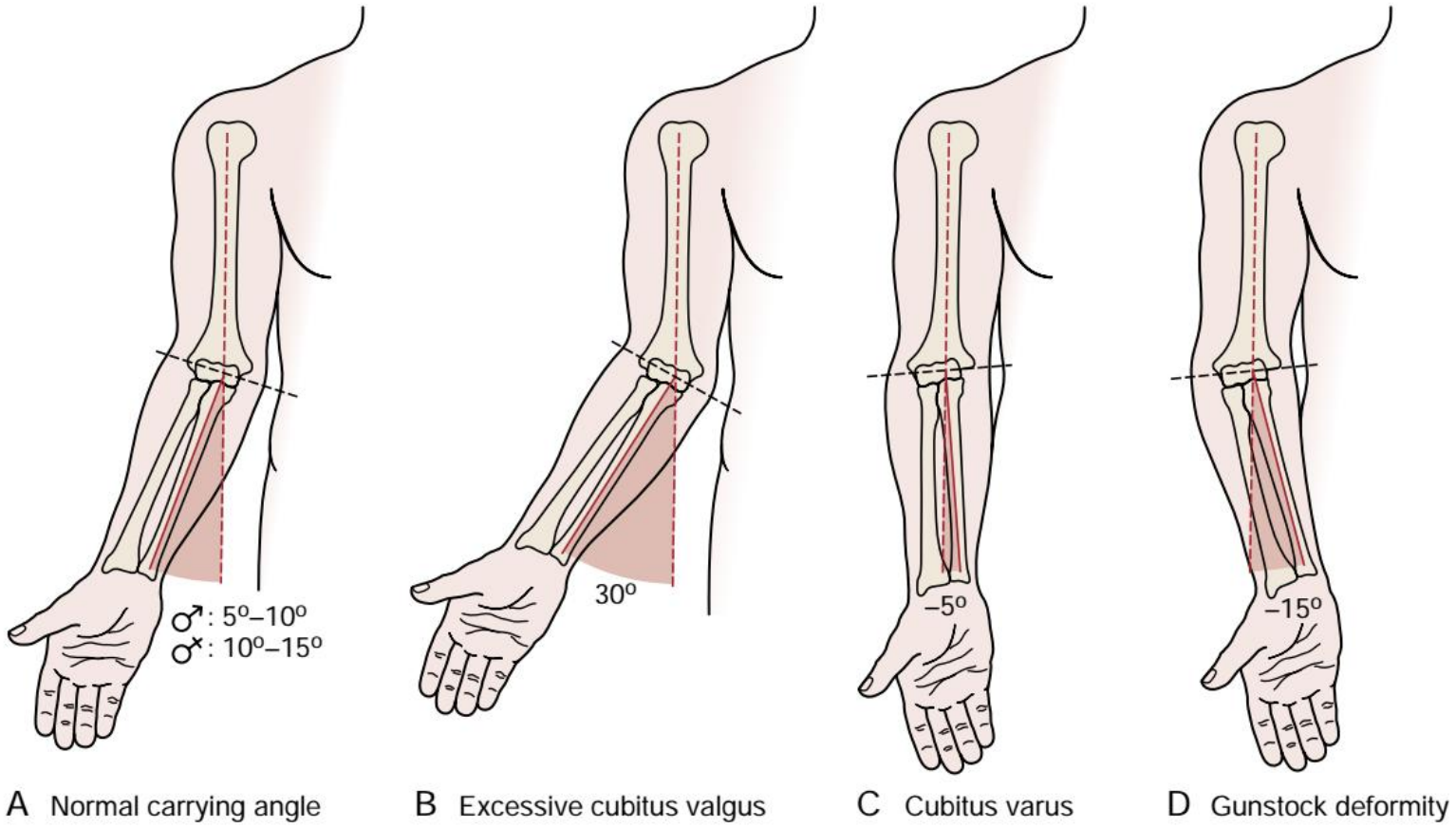








Carrying angle



A Normal carrying angle

B Excessive cubitus valgus

C Cubitus varus

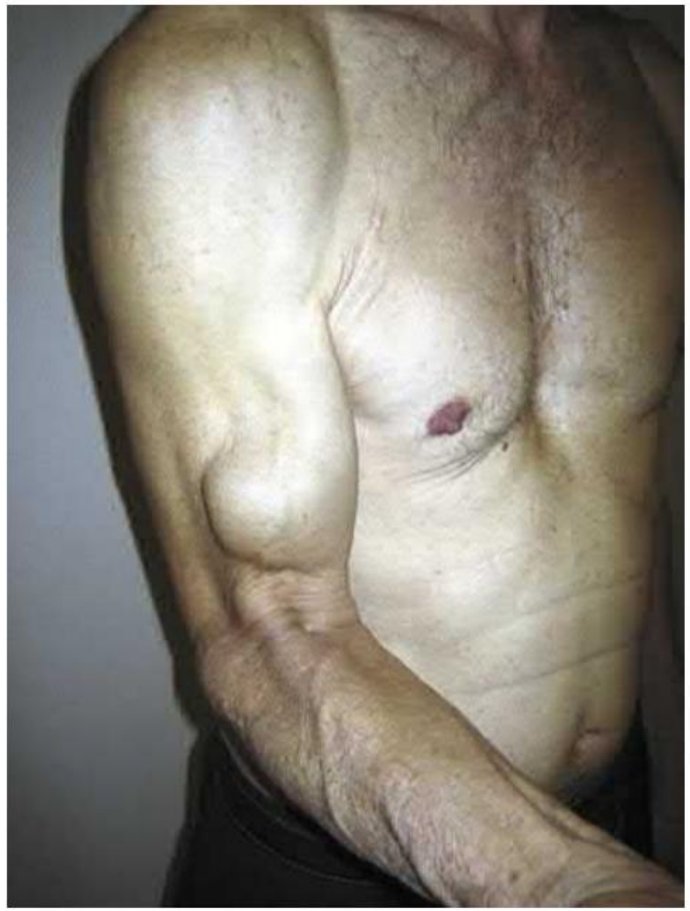
D Gunstock deformity

Hyper-flexibility, stiff elbow, olecranon bursitis and Popeye's sign









The Ehlers Danlos Society

THE BEIGHTON SCORING SYSTEM
Measuring joint hypermobility

A. 5th FINGER / 'PINKIES'

Test **both sides**: Rest palm of the hand and forearm a **flat surface** with palm side down and fingers out straight.

Can the **fifth finger** be bent/lifted upwards at the knuckle to go back **beyond 90 degrees?**

If yes, add **one point** for each hand.



1 POINT FOR EACH HAND



B. THUMBS

Test **both sides**: With the arm out straight, the palm facing down, and the wrist then fully bent downward, can the thumb be pushed back to touch the forearm?

If yes, add **one point** for each thumb.



1 POINT FOR EACH THUMB

C. ELBOWS

Test **both sides**: With arms outstretched and palms facing upwards, does the elbow extend (bend too far) upwards **more than an extra 10 degrees** beyond a normal outstretched position?

If yes, add **one point** for each side.



1 POINT FOR EACH ARM



1 POINT FOR EACH LEG

D. KNEES

Test **both sides**: While standing, with knees locked (bent backwards as far as possible), does the lower part of either leg extend **more than 10 degrees forward?**

If yes, add **one point** for each side.

E. SPINE

Bend forward, can you place the palms of your hands **flat on the floor in front of your feet without bending your knees?**

1 POINT

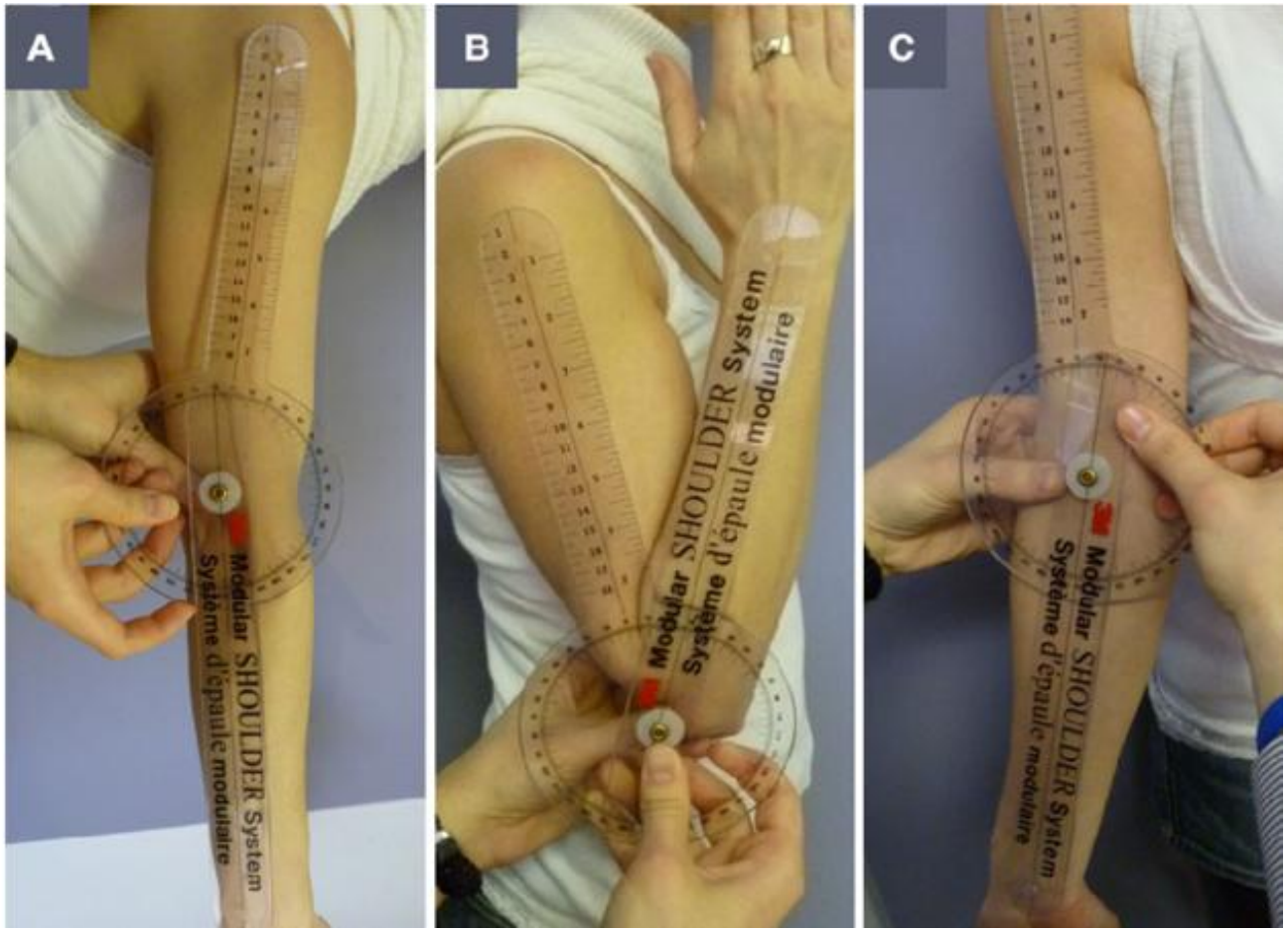


If yes, **add one point.**

ROM Assessment

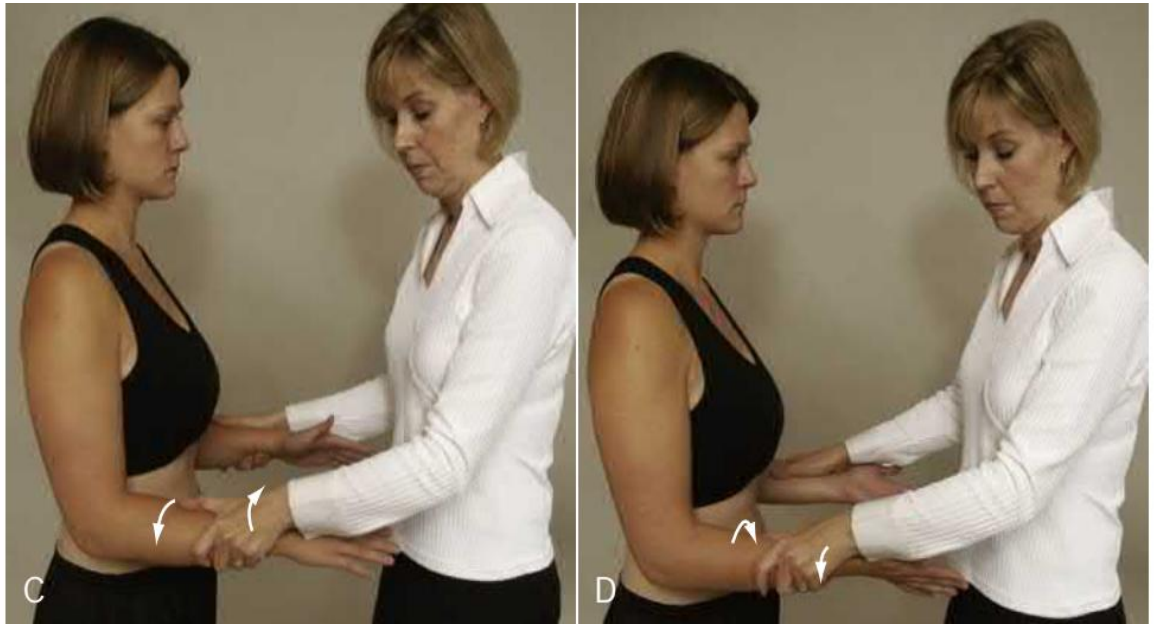


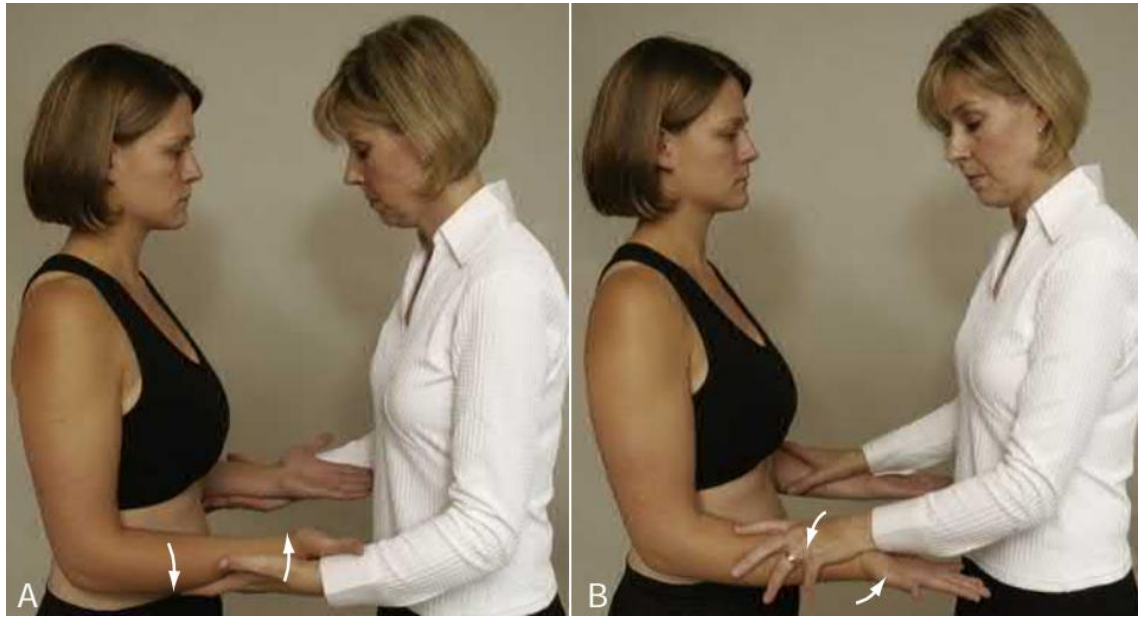
Goniometer





MRC Muscle testing





Flexibility testing



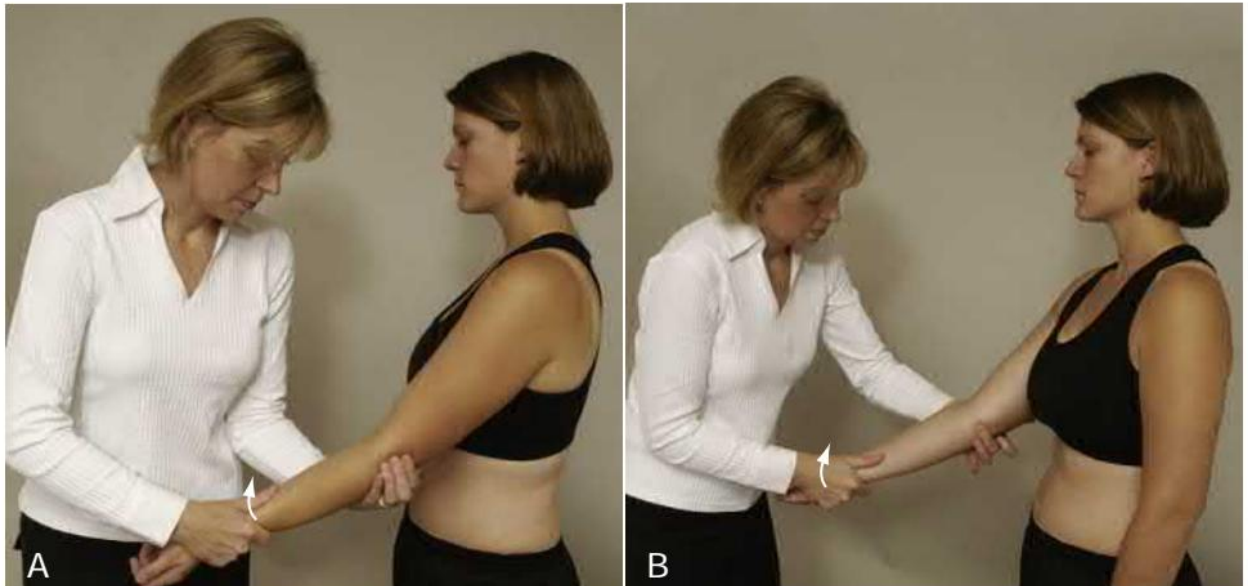
Special Tests

Elbow Joint

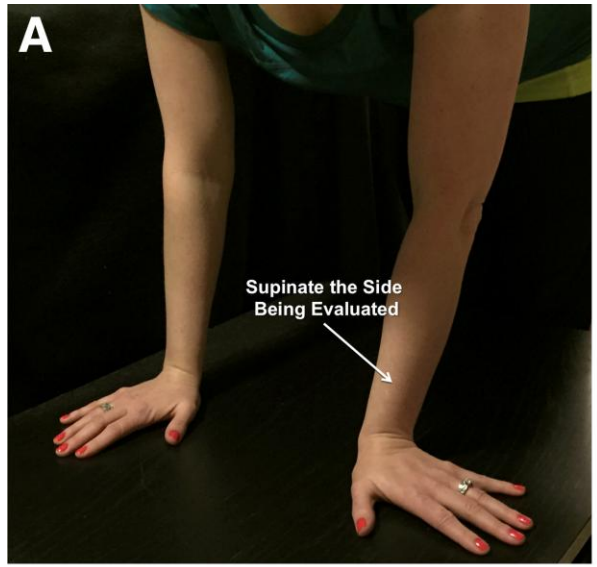
Joint Integrity testing

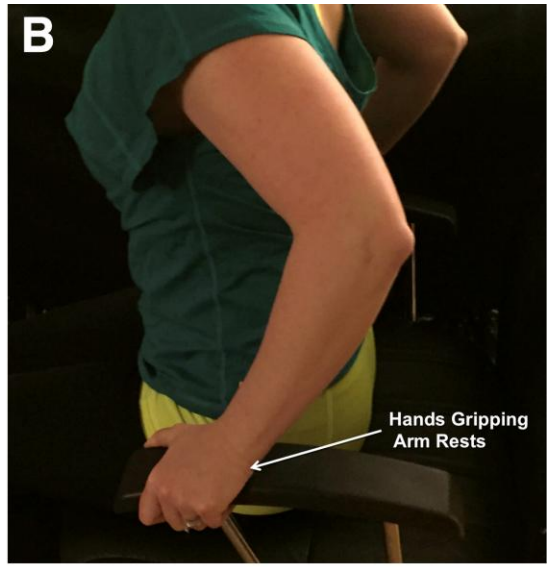


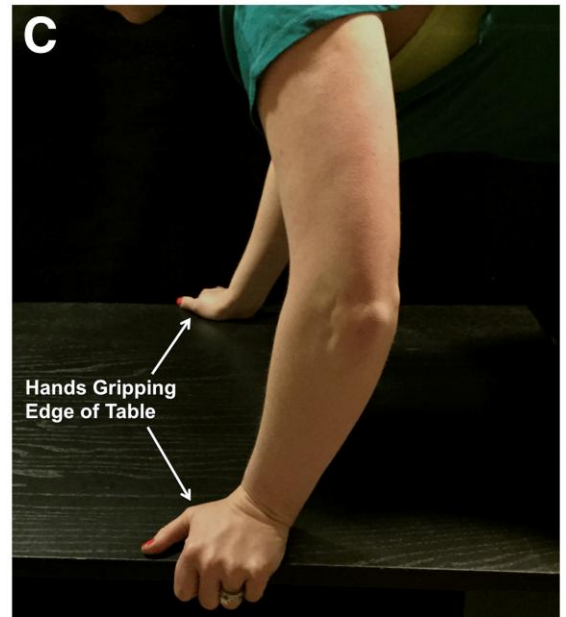
Valgus and varus stress tests



The prone push-up test (A), the chair push-up test (B), and the tabletop relocation test (C)

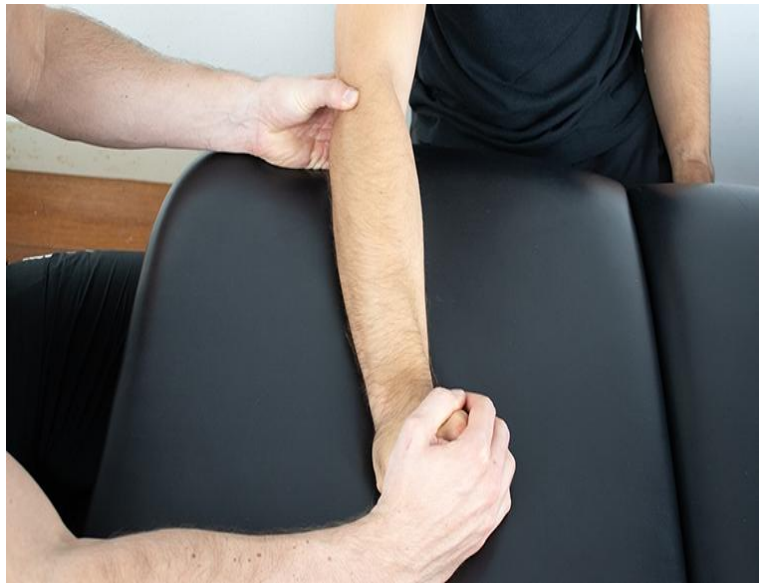






Lateral Epicondylitis

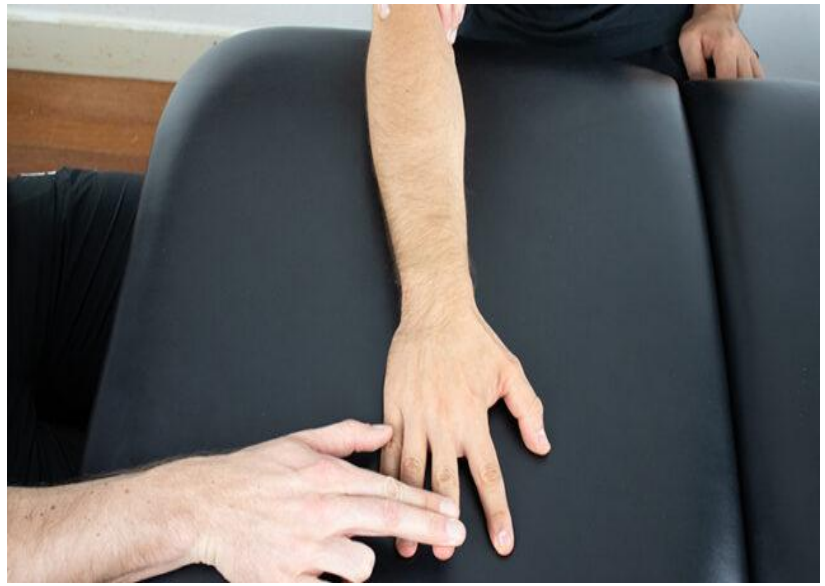
Cozen’s Test



Mille’s Test



Maudsley’s test



Medial Epicondylitis



Distal Biceps Tendon Ruptures



Supination Pronation Test

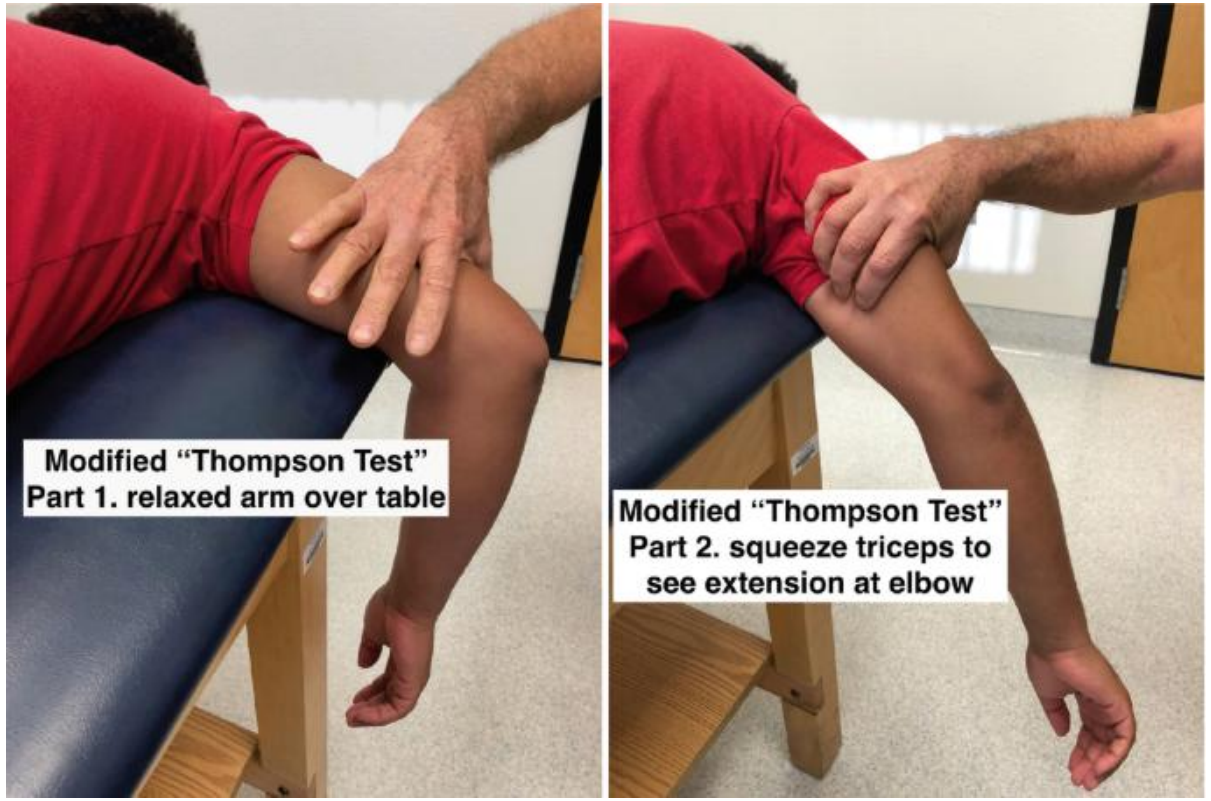


Hook Test



Triceps Tendon Rupture

Modified Thompson Test



Pronator Teres Syndrome



Cubital Tunnel Syndrome

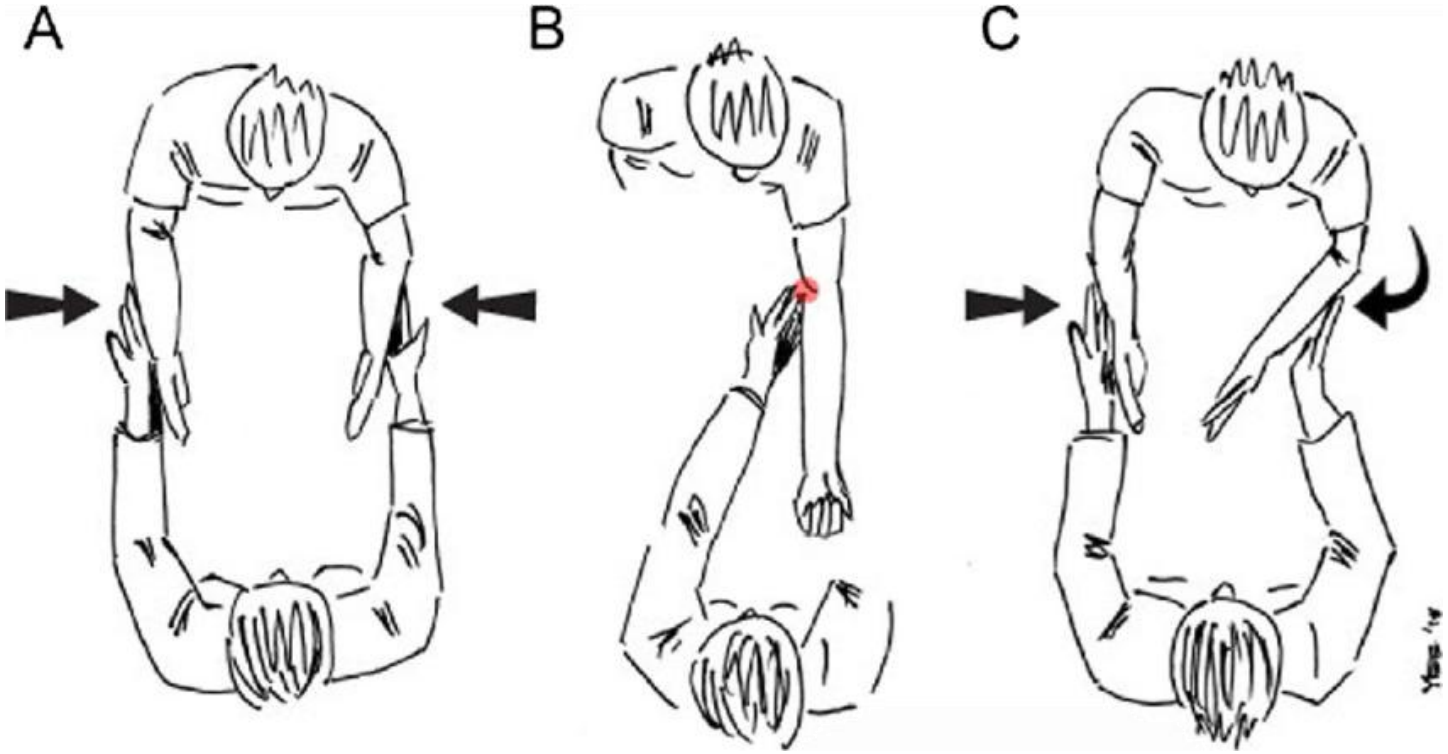
Tinel's Test



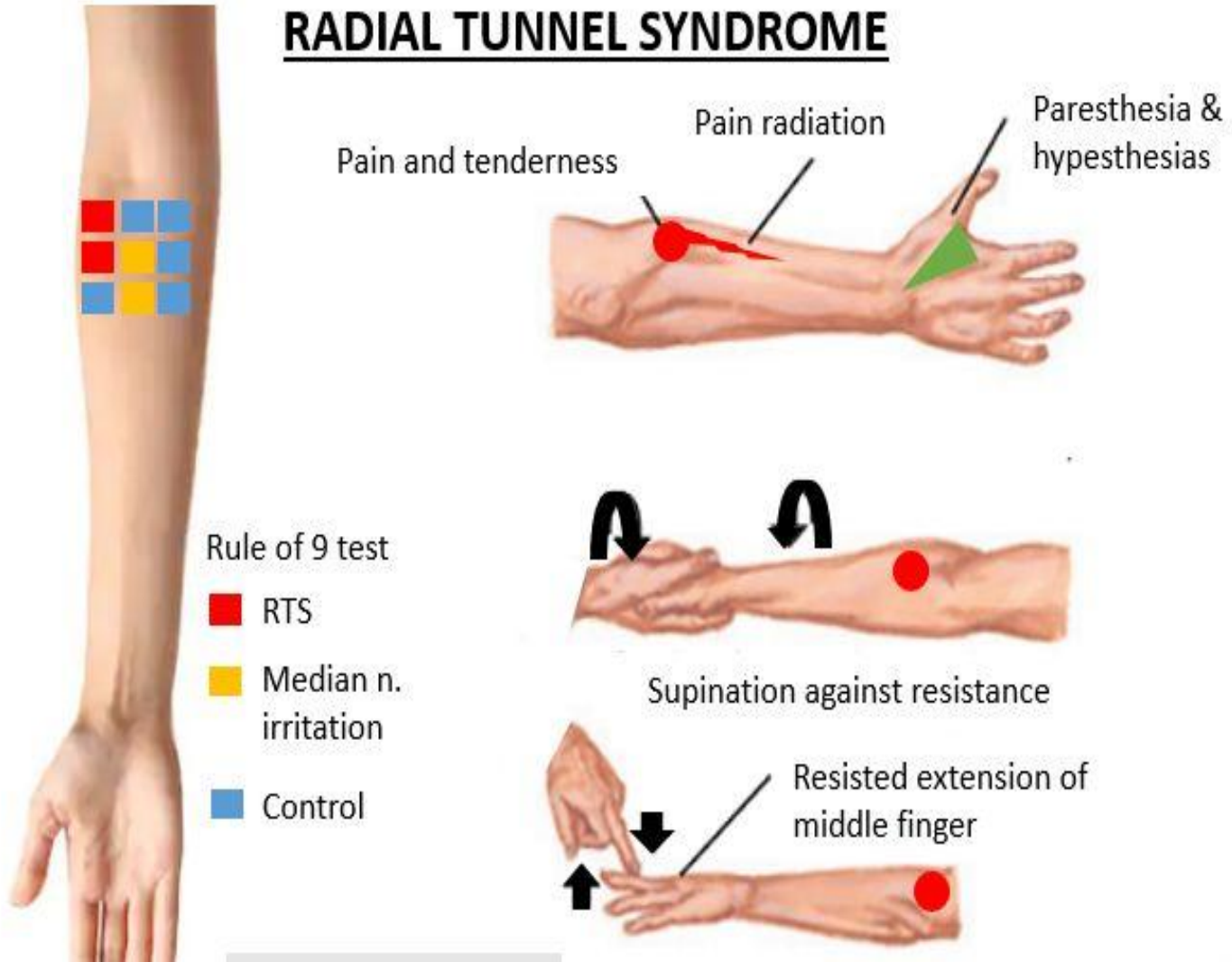
Elbow flexion test

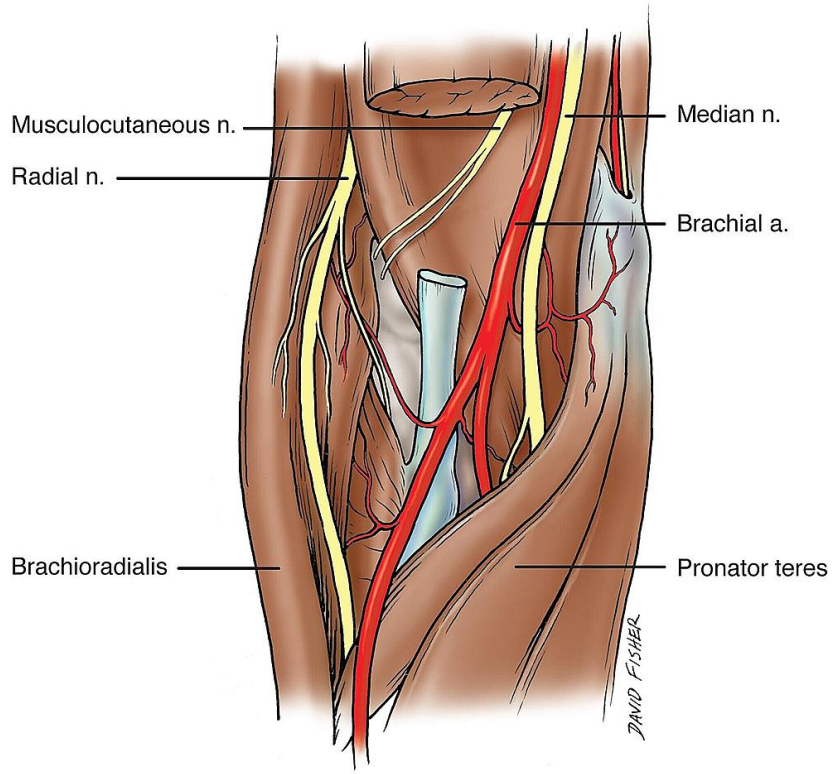


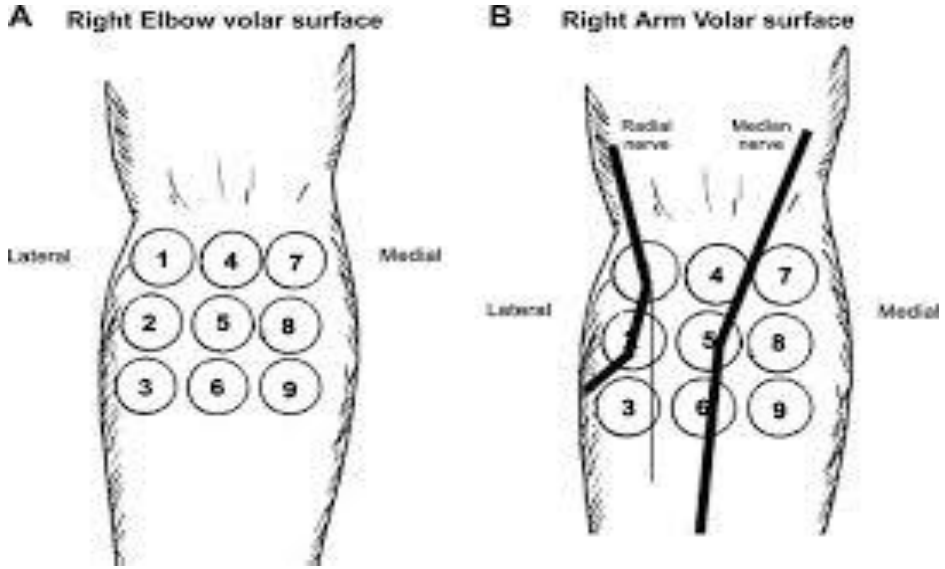
scratch collapse test



Radial Tunnel Syndrome

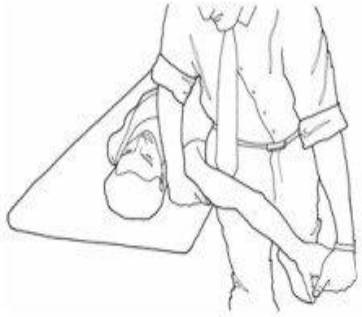






Neurological Assessment

Neuro integrity testing for neurological symptoms



ULNT$_{MEDIAN\,(1)}$



ULNT$_{MEDIAN\,(2a)}$



ULNT$_{RADIAL\,(2b)}$



ULNT$_{ULNAR\,(3)}$

Neurodynamic testing

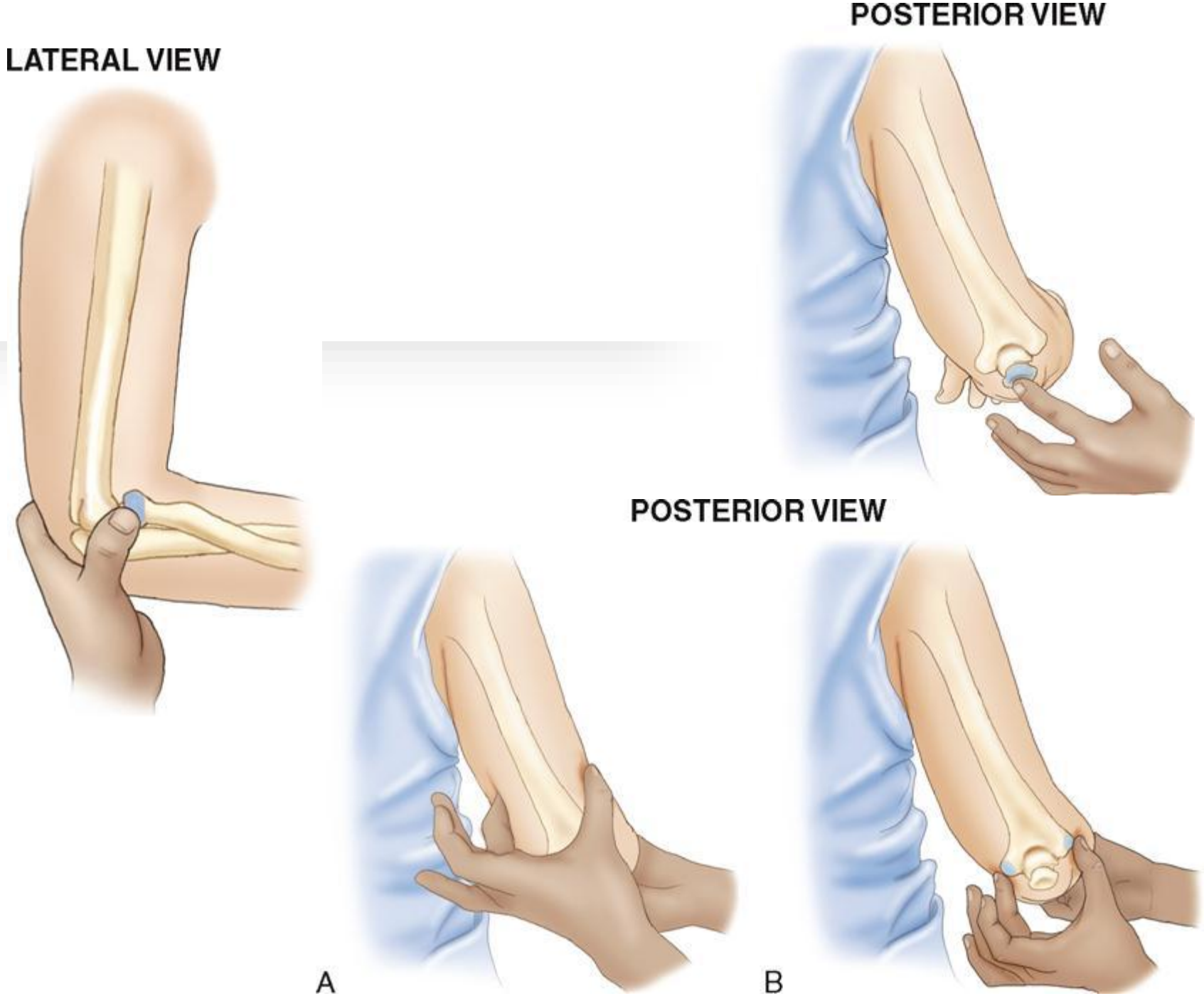
Vascular Assessment

Brachial Artery Palpation



Palpation

- Joint line
- Soft tissue
- Bursae
- Tendons
- Muscles
- Ligaments
- Bony prominences
- Pulse



Questions

